

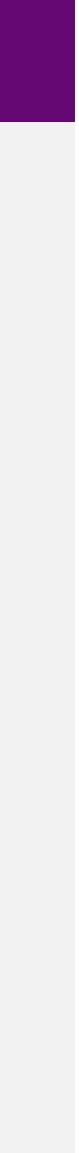


Multi-Agent Owner-Assisted Scoring Mechanisms

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Why we need owner-assisted scoring?

Single-agent
(artist/collector)
scoring.¹

HELL (1925-1992)

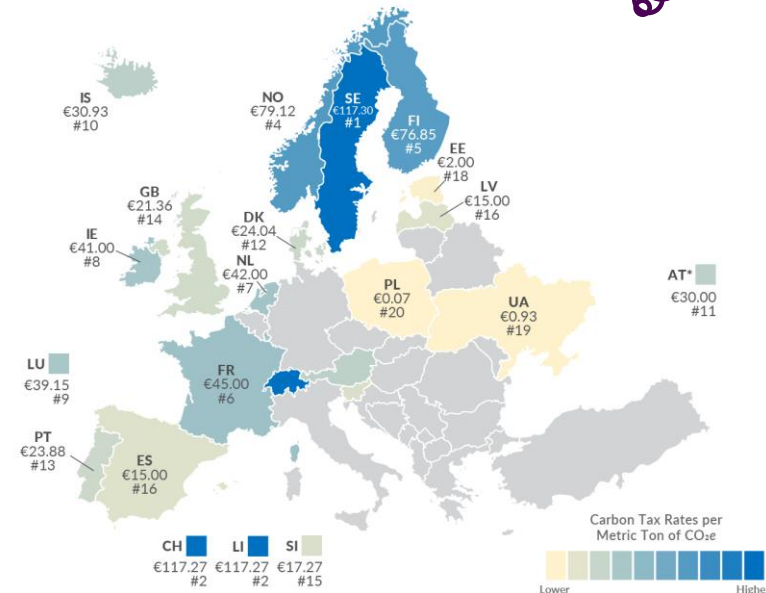


Price Realized
\$6,283,750 (Set Currency)

Estimate
\$3,000,000 - \$4,000,000

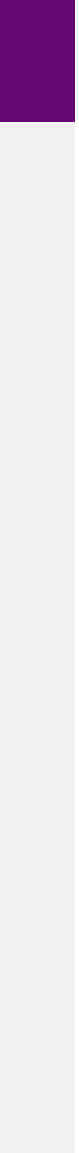
Sale Information
SALE 2785 —
POST-WAR & CONTEMPORARY
EVENING SALE
15 May 2013
New York, Rockefeller Plaza

Carbon Taxes in Europe
Carbon Tax Rates per Metric Ton of CO₂e, as of April 1, 2022



Taxing requires
information from
multiple owners.

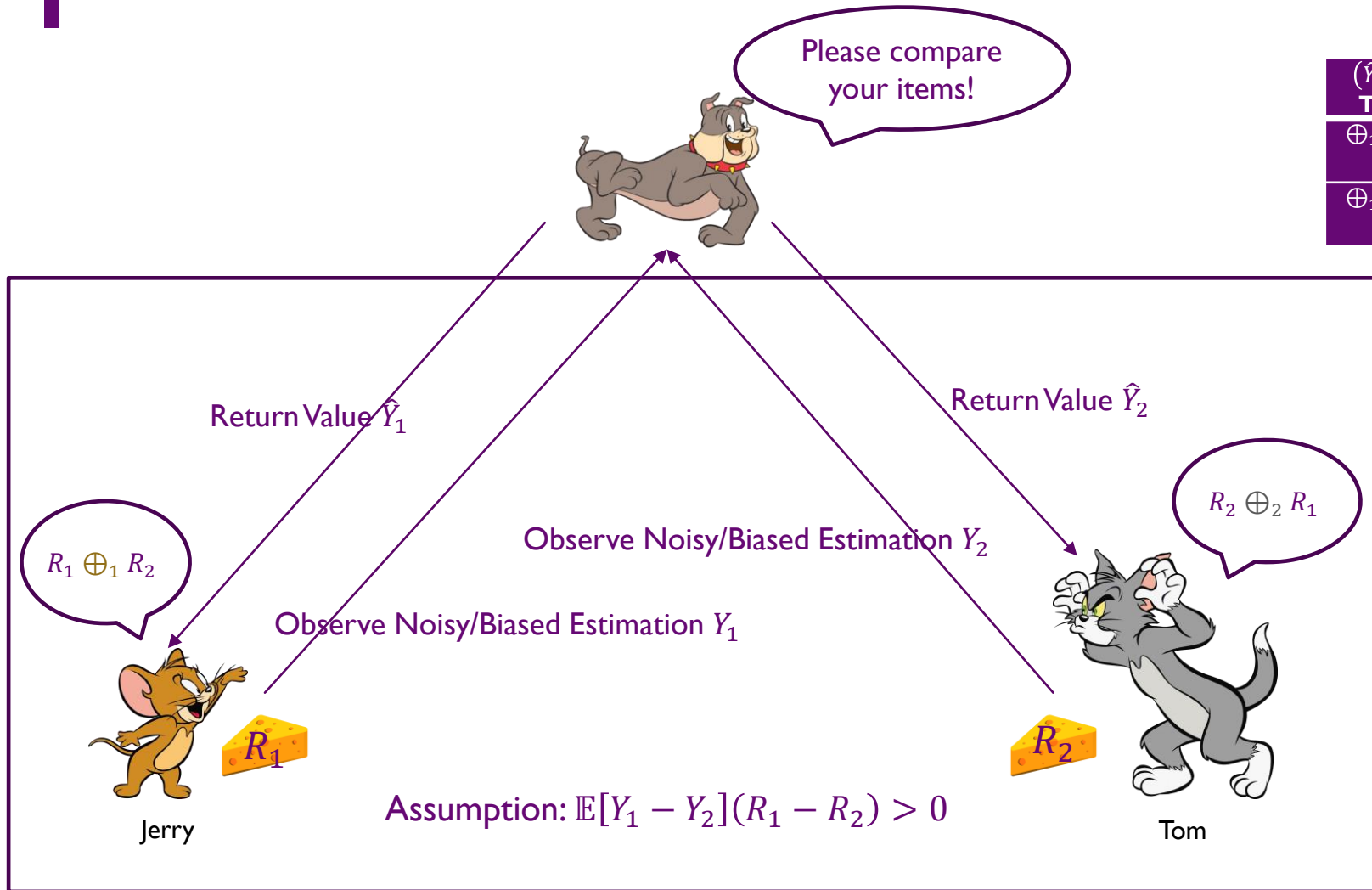
Soliciting information from owners to improve value evaluation!



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Mechanism $\mathcal{M}_2$ for Two Agents

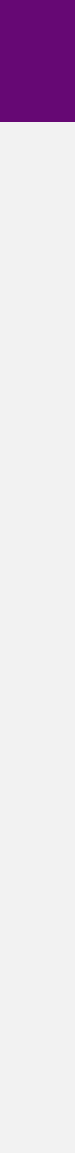


Our Mechanism Design

$(\hat{Y}_1, \hat{Y}_2)$ Table	$Y_1 \oplus_1 Y_2$	$\neg(Y_1 \oplus_1 Y_2)$
$\oplus_1 \neq \oplus_2$	(Y_1, Y_2)	$\left(\frac{Y_1 + Y_2}{2}, \frac{Y_1 + Y_2}{2}\right)$
$\oplus_1 = \oplus_2$	$\left(\frac{3Y_1 - Y_2}{2}, \frac{3Y_2 - Y_1}{2}\right)$	$\left(\frac{3Y_1 - Y_2}{2}, \frac{3Y_2 - Y_1}{2}\right)$

Satisfying two design goals for $\hat{Y}_1, \hat{Y}_2$:

- incentive-compatibility (truthful $\oplus_1$ and $\oplus_2$),
- estimation improvement ($|\hat{Y} - R|_2 \leq |Y - R|_2$ if agents are both truthful).



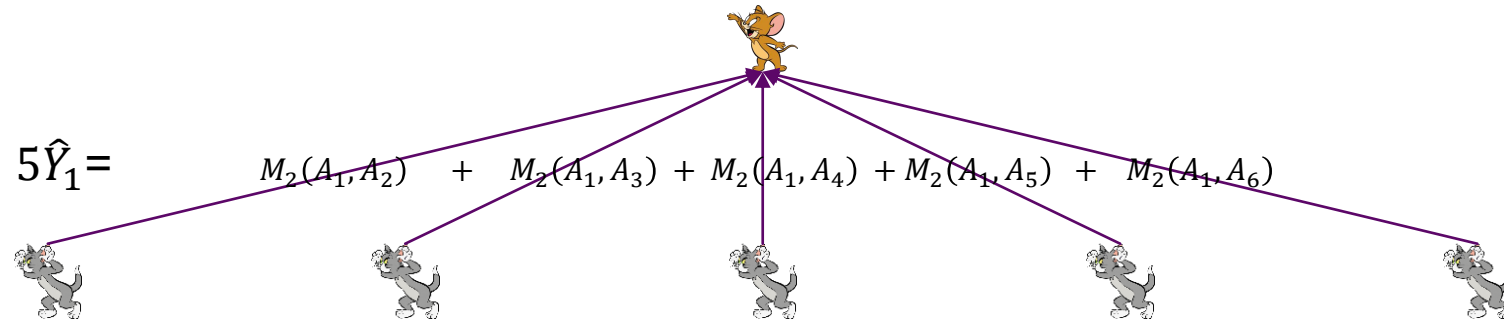
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Mechanism $\mathcal{M}_n$ for Multiple (n) Agents

Mechanism $\mathcal{M}_n$:

- Step 1: get noisy/biased estimation of $\mathbf{Y}$ for true item values $\mathbf{R}$.
- Step 2: for each pair $P_k = (A_i, A_j)$ of agents in pair set $\mathbf{P}$, perform mechanism $\mathcal{M}_2$ and get estimations $M_2(A_i, A_j)$ and $M_2(A_j, A_i)$ for those two agents, respectively.
- Step 3: return estimated value $\hat{Y}_i = \frac{\sum_{j=1, j \neq i}^n M_2(A_i, A_j)}{n-1}$ where $M_2(A_i, A_j)$ is set as $\frac{Y_i}{n-1}$ if (A_i, A_j) is not in $\mathbf{P}$.



According to linearity of expectation, one could verify that $\mathcal{M}_n$ still satisfies **incentive-compatible** and **estimation improvement**.

Several Extensions

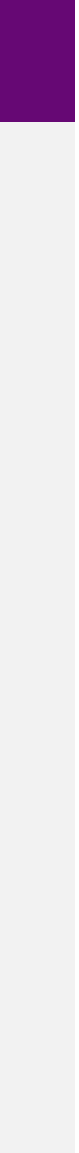
- We use the identity utility function (or linear utility function) previously. I also prove that with an assumption on Y , our mechanisms $\mathcal{M}_2$ and $\mathcal{M}_n$ still guarantee **incentive-compatible** when agents have **bilipschitz utility functions**.

$$\kappa_1|x - y| \leq |U(x) - U(y)| \leq \kappa_2|x - y|$$

- By adding a constant ϵ smaller than 0.5, new mechanisms $\mathcal{M}'_2$ and $\mathcal{M}'_n$ support strictly **incentive-compatible** for multi-agent setting while each agent has **multiple items**.

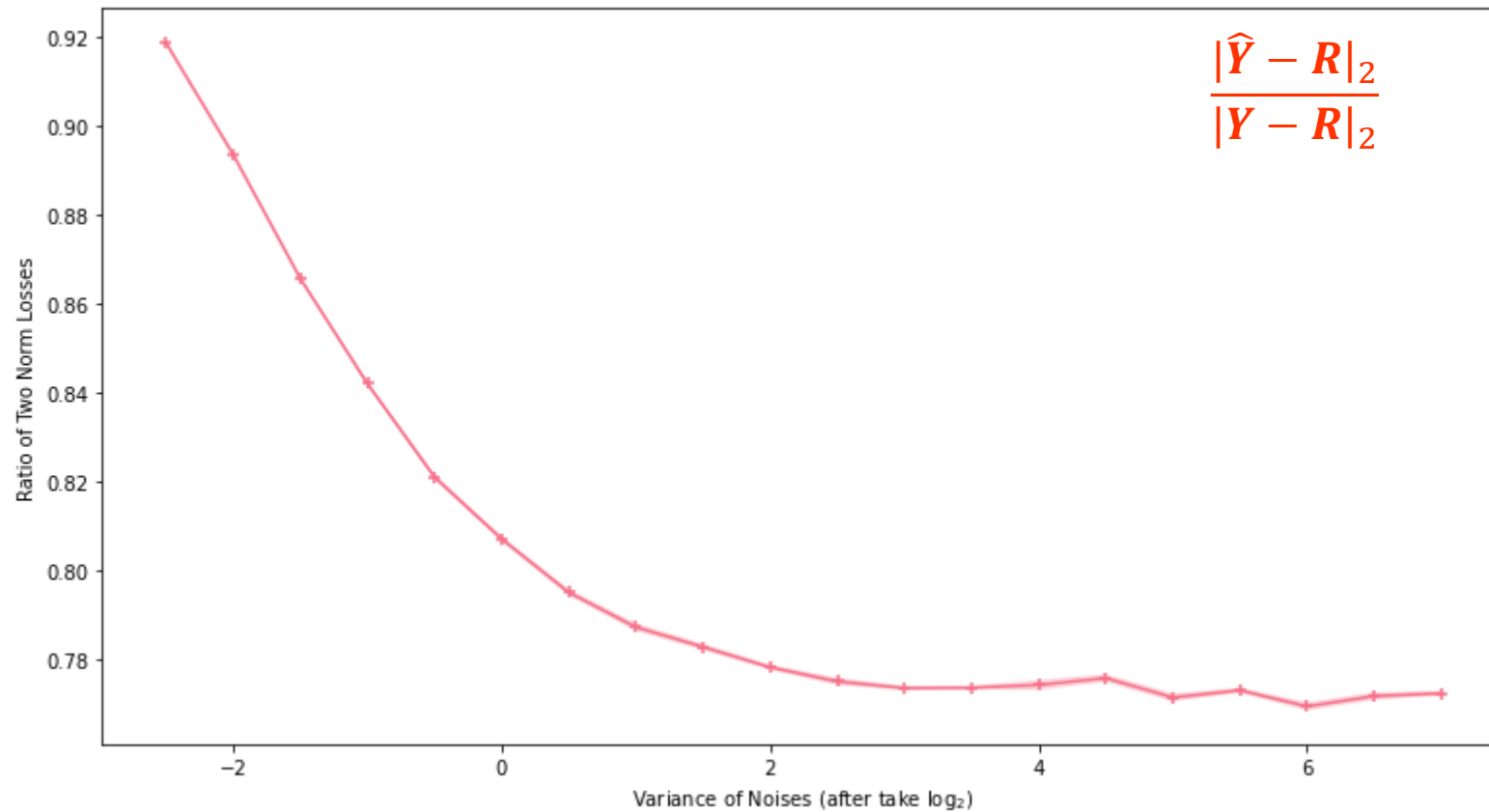
$\hat{Y}$ Design for $\mathcal{M}'_2$ and $\mathcal{M}'_n$

$(\hat{Y}_1, \hat{Y}_2)$ Table	$Y_1 \oplus_1 Y_2$	$\neg(Y_1 \oplus_1 Y_2)$
$\oplus_1 \neq \oplus_2$	(Y_1, Y_2)	$\left(\frac{Y_1 + Y_2}{2} + \epsilon \frac{Y_1 - Y_2}{2} \mathbb{I}[\oplus_1 = >] + \epsilon \frac{Y_2 - Y_1}{2} \mathbb{I}[\oplus_1 = <], \frac{Y_1 + Y_2}{2} + \epsilon \frac{Y_2 - Y_1}{2} \mathbb{I}[\oplus_2 = >] + \epsilon \frac{Y_1 - Y_2}{2} \mathbb{I}[\oplus_2 = <] \right)$
$\oplus_1 = \oplus_2$	$\left(\frac{3Y_1 - Y_2}{2}, \frac{3Y_2 - Y_1}{2} \right)$	$\left(\frac{3Y_1 - Y_2}{2}, \frac{3Y_2 - Y_1}{2} \right)$



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Simulations

We run $\mathcal{M}_n$ for all pairs of agents on random item values R and standard distribution noises ϵ such that $Y = R + \epsilon$. $\hat{Y}$ generated by $\mathcal{M}_n$ achieves **8%-20% improvement** over Y .



Q&A

Thank you for your attention!